

Data Detective Final Project

Trade Data Validation, Reporting & Reconciliation

Objective: Validate daily trading activity, identify data quality issues, reconcile position changes, and communicate operational discrepancies.

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Summary of Workflow

End-to-End Trade Validation Workflow

- Raw Data → Data Inspection → Issue Detection → Data Cleaning → Reporting → Reconciliation → Dashboard

Raw Data Summary

Dataset	Size (rows)
Trades (sample_trades.csv)	286
Position T0 (positions_t0.csv)	31
Position T1 (positions_t1.csv)	30
Corporate Actions (corp_actions.csv)	9

Data Cleaning & Issues Identified

Overall Issues Detected

Null values per column:	
trade_id	0
symbol	0
account	0
trade_date	0
side	0
quantity	0
price	4
venue	4
trade_type	0
signed_qty	0
notional	4
fee	0
settlement_date	0

Duplicate trade IDs:
8
Zero quantity trades:
6
Negative quantity trades:
6
Settlement date issues:
6

Cleaning Result

- 286 → 253 rows
- 33 records removed

3 key Issues & Cleaning Actions

Issue	Cleaning Action	Removed
Duplicate IDs	Kept first record per trade_id	8
Negative quantity	Removed invalid quantity rows	6
Missing price	Removed incomplete records	4

Duplicate trade IDs:							
	trade_id	symbol	account	trade_date	side	quantity	price
99	T100000	META	ACC2	2023-03-04	SELL	3729	696.7500
101	T100000	META	ACC2	2023-03-04	SELL	3772	6.9634
229	T100012	GOOGL	ACC2	2023-03-09	BUY	4116	699.3100
232	T100012	GOOGL	ACC2	2023-03-09	BUY	4084	705.6100
Negative quantity trades:							
	trade_id	symbol	account	trade_date	side	quantity	price
0	T100165	AAPL	ACC1	2023-03-01	SELL	-4851	910.53
4	T100248	AMZN	ACC2	2023-03-01	BUY	-4511	308.28
30	T100069	XOM	ACC1	2023-03-01	BUY	-484	477.96
242	T100031	NVDA	ACC1	2023-03-09	BUY	-3541	651.66
251	T100229	XOM	ACC3	2023-03-09	BUY	-1396	453.45
285	T100024	XOM	ACC2	2023-03-10	SELL	-2198	132.98
Missing price:							
	trade_id	symbol	account	trade_date	side	quantity	price
98	T100212	META	ACC1	2023-03-04	BUY	4062	NaN
143	T100077	XOM	ACC2	2023-03-05	SELL	4130	NaN
149	T100173	AAPL	ACC2	2023-03-06	BUY	1289	NaN
158	T100088	GOOGL	ACC2	2023-03-06	BUY	4954	NaN

Reporting Results & Observations

Reports Generated

- cleaned_trades.csv
- daily_report.csv
- symbol_report.csv

Key Observations

- AAPL had the highest total trading volume: **88,307 shares**
- META had the largest positive net quantity: **+31,813**
- MSFT had the largest negative net quantity: **-8,691**

symbol_report					
symbol	trade_count	total_quantity	net_qty	total_notional	avg_price
AAPL	32	88307	11329	45834786.8333	503.027465625
AMZN	26	58215	5861	29592730.32	557.065
GOOGL	29	77946	666	43328126.1157	558.0165413793100
JPM	21	57319	-127	29349113.9	474.27476190476200
META	29	75487	31813	46506061.0624	598.6802620689660
MSFT	23	63895	-8691	36576825.5484	578.2423652173910
NVDA	30	68594	-928	27634077.91	446.0776666666670
SPY	20	53425	577	22728917.52	403.498
TSLA	20	48666	-7390	22439494.0258	453.806765
XOM	23	67511	-7177	35834440.45	519.2121739130430

Reconciliation Logic

Actual Position Change

- $\Delta \text{ Position} = \text{Position T1} - \text{Position T0}$

Expected Position Change

- $\text{Expected } \Delta = \text{Net Trade Quantity} + \text{Corporate Action Shares}$

Reconciliation Difference

- $\text{Recon Diff} = \Delta \text{ Position} - \text{Expected } \Delta$
- $\text{Recon Diff} = 0 \rightarrow \text{MATCH}$
- $\text{Recon Diff} \neq 0 \rightarrow \text{MISMATCH}$

Example

symbol	position_t0	position_t1	net_qty	corp_shares	corp_action_missing	delta_pos	expected_delta	recon_diff	status
AAPL	18414	26181	11329.0	0.0	FALSE	7767	11329.0	-3562.0	MISMATCH
MSFT	9693	1002	-8691.0	0.0	TRUE	-8691	-8691.0	0.0	MATCH

Mismatch Case Study 1: GOOGL Missing Position Data

Issue Detected

```
Missing position rows:
  symbol account  position_t0  position_t1    _merge
7  GOOGL   ACC2         8742         NaN  left_only
```

symbol	position_t0	position_t1	net_qty	corp_shares	corp_action_missing	delta_pos	expected_delta	recon_diff	status
GOOGL	22902	9149	666.0	0.0	FALSE	-13753	666.0	-14419.0	MISMATCH

Root cause

- ACC2 has a T0 position of 8,742 shares but no corresponding T1 position record.
- The missing T1 record is a major position-data integrity issue and makes the GOOGL reconciliation unreliable.

Mismatch Case Study 2 – JPM Corporate Action Issue

Issue Detected

```
Missing corporate action shares:
  symbol    ex_date corp_type  corp_shares
8    JPM 2023-03-01      DIV        NaN
```

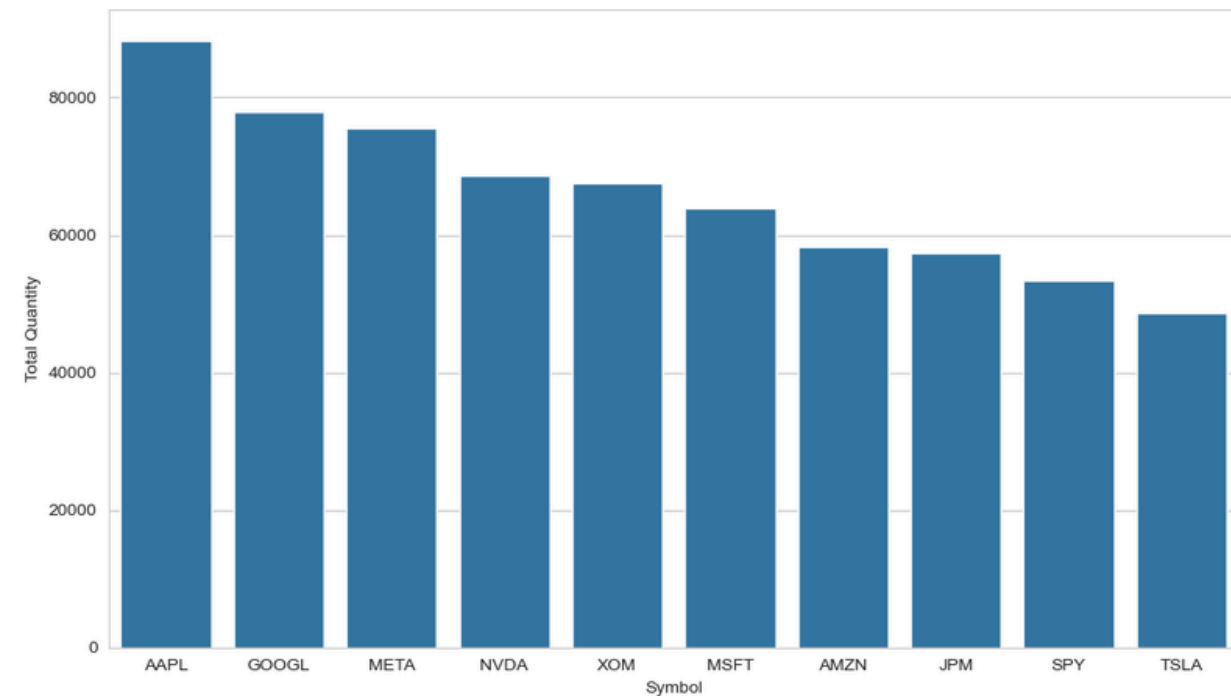
symbol	position_t0	position_t1	net_qty	corp_shares	corp_action_missing	delta_pos	expected_delta	recon_diff	status
JPM	18553	14619	-127.0	0.0	TRUE	-3934	-127.0	-3807.0	MISMATCH

Root cause

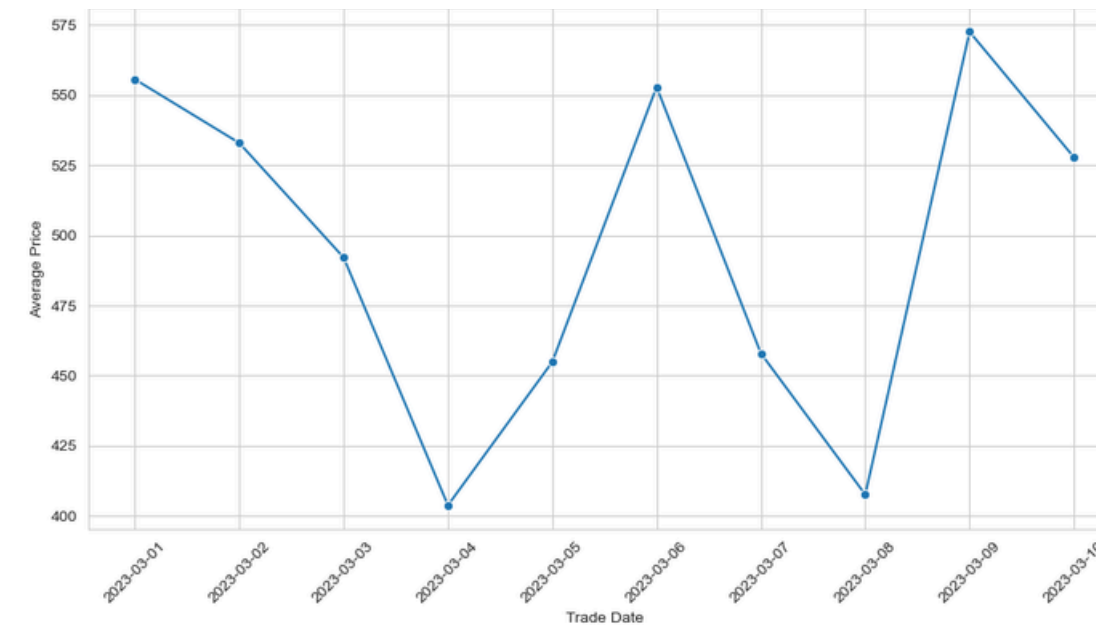
- The corporate-action input is incomplete, so the expected position movement may be incorrect.

Dashboard Highlights & Insights

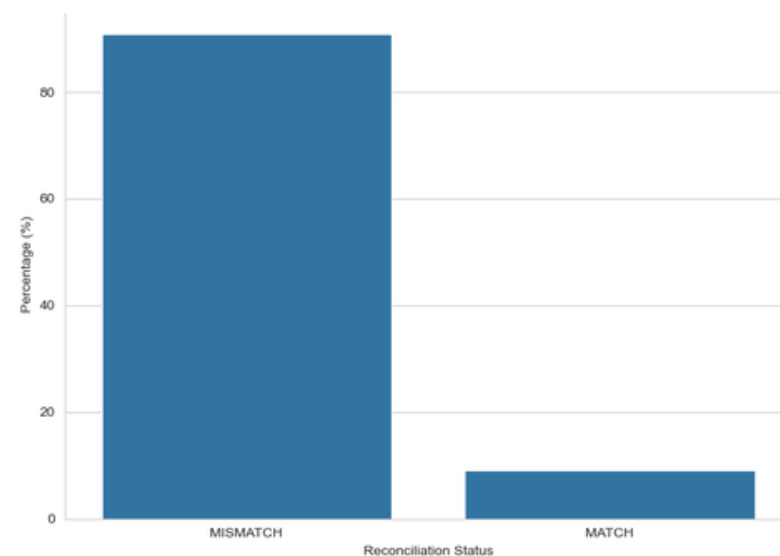
Which symbols had the highest trading volume?



How did AAPL's price evolve over time?



What % of positions reconciled correctly?



- **AAPL** had the highest trading volume with 88,307 shares.
- **AAPL's** average price decreased from **\$555.62** to **\$527.86** over the analyzed period.
- Only 9.1% of positions reconciled correctly (1 of 11 symbols)

Lessons Learned & Conclusion

Technical Lessons

- Data quality before analytics

- Validate before aggregating or visualizing
- Recalculate derived fields after cleaning
- Use multiple datasets to validate results

Operational Lessons

- A mismatch is a starting point, not a conclusion

- Trade errors can affect downstream positions
- Missing position records can create large discrepancies
- Corporate-action data must be validated separately
- Exceptions require root-cause investigation

Conclusion

- Reliable reconciliation depends on both clean trade data and complete reference/position data.



Thank you

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